

With increasing number of patients and the cardiac related problems, visits are becoming a heavy burden on healthcare systems. Remote monitoring communicates the information from the implanted device without the patient physically visiting the centre while the patient stays connected to the healthcare centre all the time. In remote monitoring, the data collected is synchronised and shared by the device with a transmitter located near the patient. The transmitter communicates the data online to a secure portal accessible by the healthcare team.

The evolution of remote monitoring technology

The first pacemaker was implanted in a human in 1958. The pacemaker did not last long but the patient lived to the age of 88 with as many as 26 pacemakers replaced one after the other in his lifetime. The modern era began in 1969 with the development of demand pacemakers, which generated a regular electrical pulse without variation on its own, on sensing that the heart was not beating regularly.

Initially, the pacemakers were able to pace only the ventricle. Next, the dual-chamber pacing, called physiologic pacing, was developed that could pace both the atrium and the ventricle. These pacemakers paced at a fixed rate without increasing the heart rate with the increased activity. The present-day 'rate responsive pacemakers' are able to increase the heart rate with increased activity.

Later, other implantable cardiovascular devices were developed to restore normal cardiac activation sequence and/or detect and treat a malignant arrhythmia. The need for remote monitoring was felt when the follow-up procedure became overwhelmingly resource-consuming and demanded differentiated human and technical resources. Moreover,

Implanted Electronic Cardiac Devices (IECDs)

A pacemaker is a small device that is placed under the skin in the chest area to help control heartbeats. It helps the heart beat more regularly if the patient has an irregular heartbeat (arrhythmia), particularly a slow one. Pacemakers use low-energy electrical pulses to control the rate and rhythm of heartbeat. Traditional pacemakers have three main parts—a pulse generator to create the electrical pulses, wires (also called leads) to carry the pulses to heart, and electrodes to sense natural heartbeat. When the heartbeat is slower than normal, the electrodes deliver electrical impulses to the heart to make it beat normally.

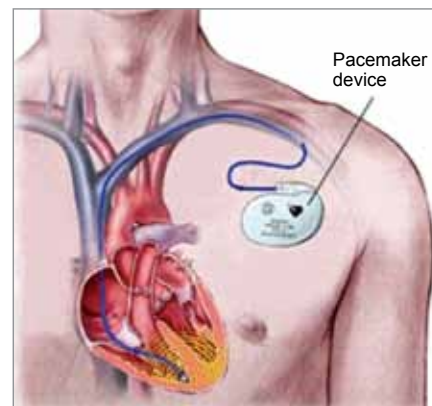


Fig. 2: Pacemaker (Source: <http://caifl.com>)

An ICD, like a pacemaker, is a small battery-powered device that monitors the heart rhythm and detects irregular heartbeats. If an abnormal heart rhythm (heart beating chaotically or too fast) is detected, the device delivers an electric shock via one or more wires connected to the heart to restore a normal heart rhythm. ICDs help in preventing cardiac arrest in high-risk patients who are at risk for life-threatening ventricular arrhythmias. Newer-generation ICDs may have a dual function, which includes the ability to serve as a pacemaker.

Cardiac resynchronisation therapy (CRT) devices. A CRT system consists of a pulse generator and thin insulated leads. The device delivers tiny amounts of electrical energy to the heart through these leads to restore normal timing of the heartbeats. This results in both ventricles pumping together and more efficiently.

There are two types of CRT devices. One is a special kind of pacemaker, called cardiac resynchronisation therapy pacemaker (CRT-P). CRT-P coordinates the timing of the upper heart chambers (atria) and the lower heart chambers (ventricles). The device monitors the heart's rhythm, detects the irregularities, and corrects them with electrical pulses. The other, called cardiac resynchronisation therapy defibrillator (CRT-D), is the same as CRT-P but includes a built-in cardioverter defibrillator (ICD) to deliver small electrical pulses to the left and right ventricles to help them contract at the same time in a more synchronised manner.

Implantable loop recorder (ILR). An ILR is a device, somewhat like a pacemaker, that records electrical activity in a patient's heart. One of the main differences is that the ILR does not regulate heart rate. Rather, it is a single-lead ECG monitoring device used for diagnosis in patients with recurrent unexplained episodes of palpitations or syncope. The device stores ECG data automatically in response to a significant bradyarrhythmias or tachyarrhythmia, or in response to patient activation. It is mostly used when symptoms are infrequent or when aggregate long-term data is required.

Implantable hemodynamic monitors (IHMs). An IHM is used as means of measuring intra-cardiac pressures and understanding the nuances of the hemodynamic derangements of this condition for accurately assessing and monitoring filling pressures to more precisely adjust therapy. IHMs also assist in management of other cardiovascular diseases, such as pulmonary arterial hypertension.

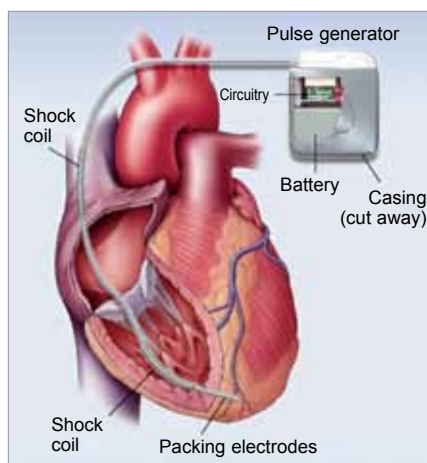


Fig. 3: Implantable cardioverter defibrillator (ICD) (Source: www.nejm.org)